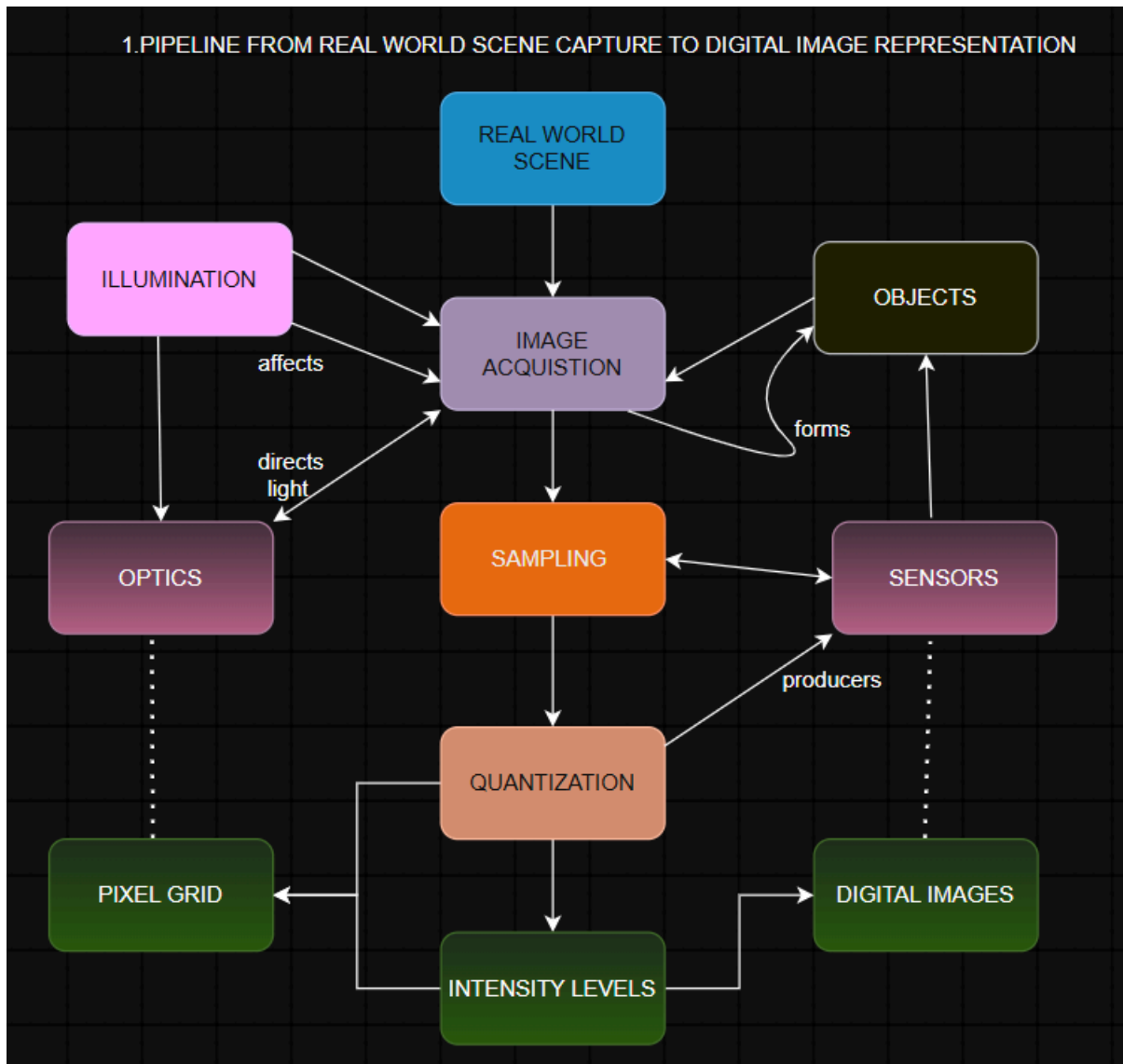


ITA0515 COMPUTER VISION

K.Amrutha

192425366

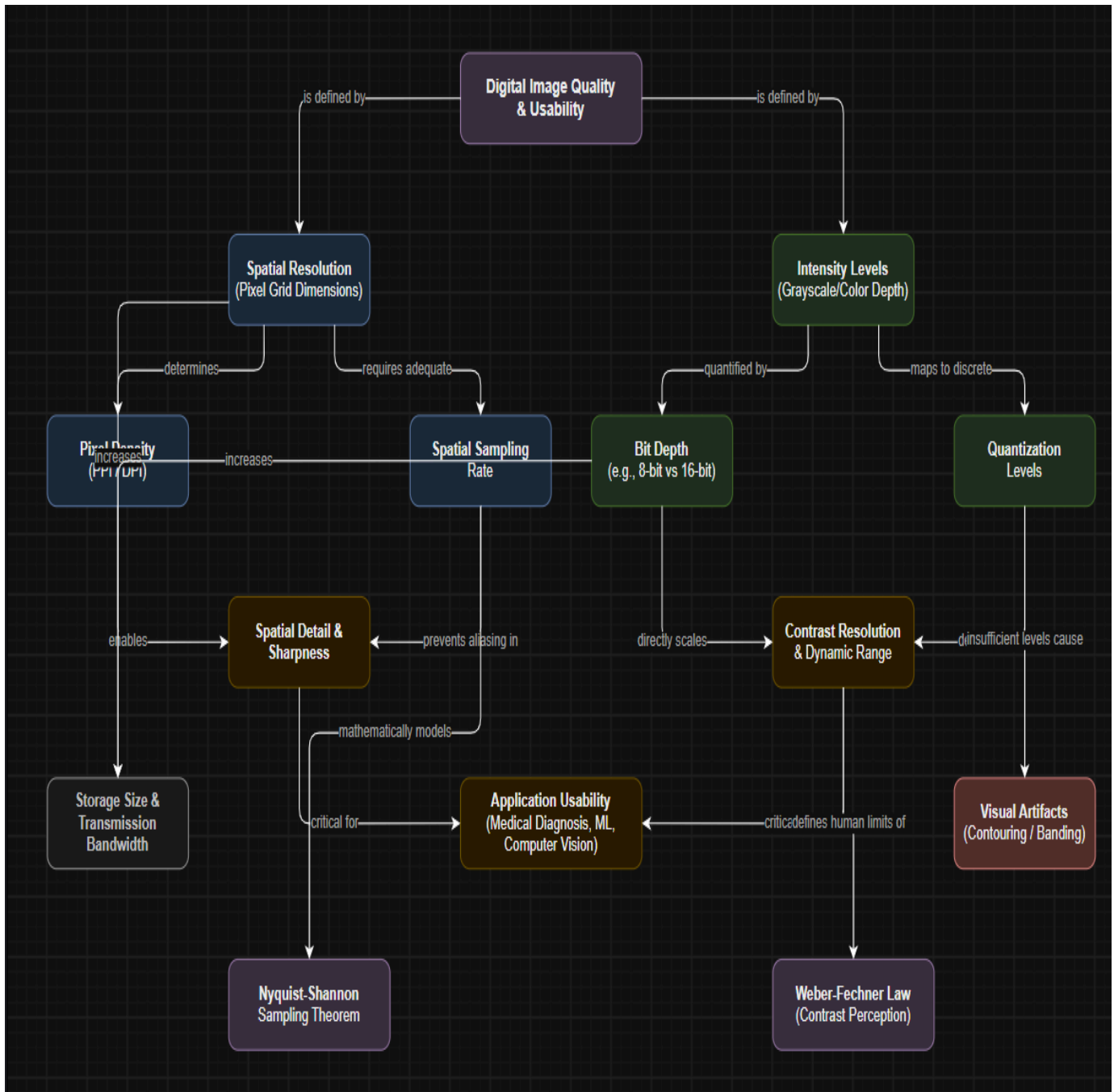
Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner



2) Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability



3) Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause–effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format



```

graph TD
    subgraph Image_Acquisition_Conditions [Image Acquisition Conditions]
        MB[Motion & Blur]
        IG[Illumination Geometry]
        AME[Atmospheric/Medium Effects]
        SC[Sensor Characteristics]
    end

    subgraph Processing_Pipeline [Processing Pipeline]
        MB --> FE[Feature Extraction]
        IG --> FE
        AME --> PD[Preprocessing & Denoising]
        SC --> PD
        FE --> PD
        PD --> NC[Normalization & Calibration]
    end

    subgraph Application_Performance [Application Performance]
        RD[Robust Descriptors] --> SP[Segmentation Precision]
        CD1[Cleaned Data] --> SP
        CD2[Cleaned Data] --> ODA[Object Detection Accuracy]
        AP[Application Performance] --> ODA
        NC --> CL[Computational Latency]
    end

    FE --> RD
    PD --> CD1
    PD --> CD2
    PD --> AP
    NC --> CL

```

The flowchart illustrates the relationship between image acquisition conditions, the processing pipeline, and application performance. It is organized into three main sections: Image Acquisition Conditions, Processing Pipeline, and Application Performance.

Image Acquisition Conditions (Top Section):

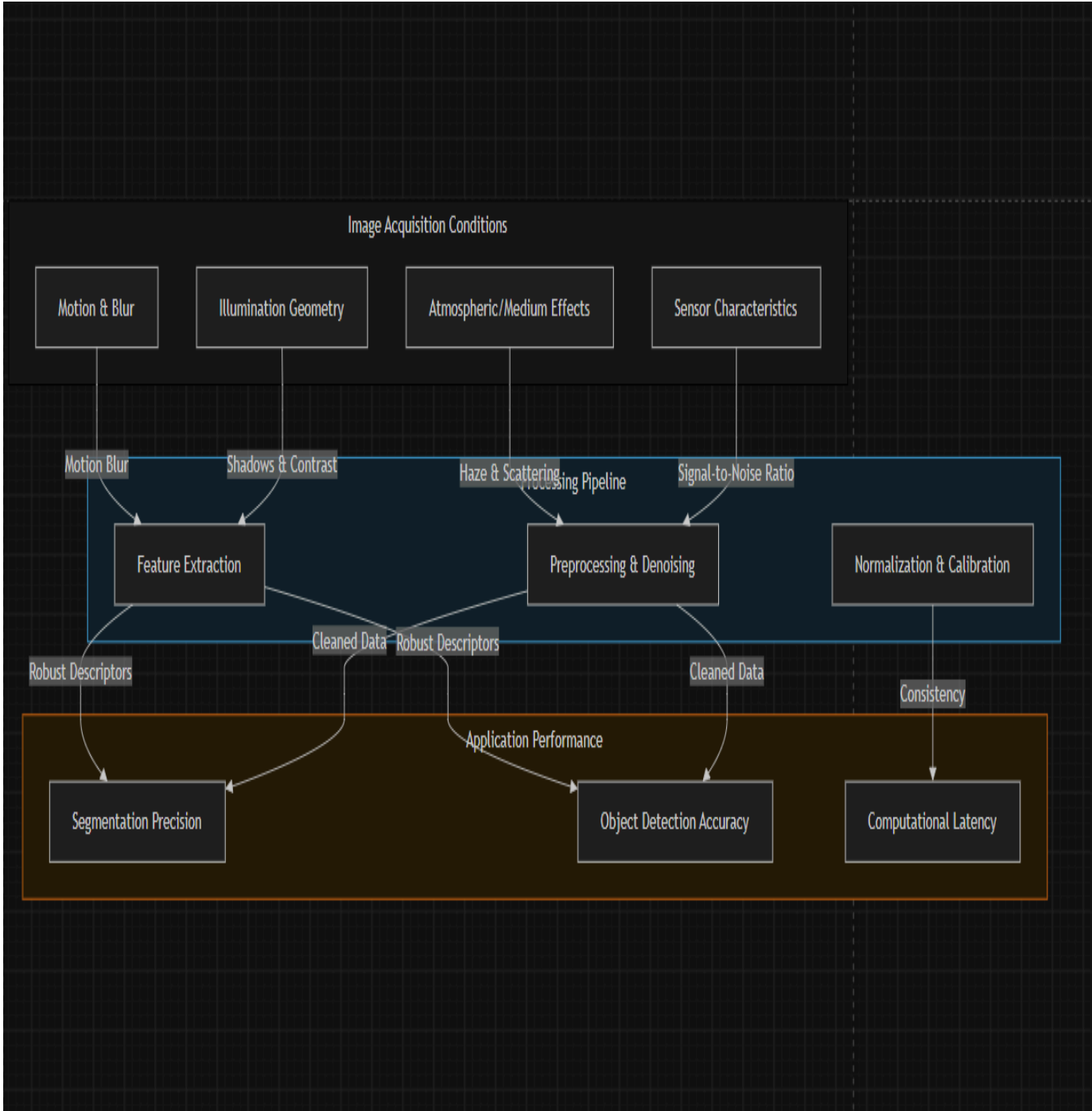
- Motion & Blur
- Illumination Geometry
- Atmospheric/Medium Effects
- Sensor Characteristics

Processing Pipeline (Middle Section):

- Feature Extraction**: Receives input from Motion & Blur and Illumination Geometry.
- Preprocessing & Denoising**: Receives input from Atmospheric/Medium Effects, Sensor Characteristics, and Feature Extraction.
- Normalization & Calibration**: Receives input from Preprocessing & Denoising.

Application Performance (Bottom Section):

- Robust Descriptors**: Receives input from Feature Extraction.
- Cleaned Data**: Receives input from Preprocessing & Denoising.
- Application Performance**: Receives input from Preprocessing & Denoising.
- Segmentation Precision**: Receives input from Robust Descriptors and Cleaned Data.
- Object Detection Accuracy**: Receives input from Cleaned Data and Application Performance.
- Computational Latency**: Receives input from Normalization & Calibration.



5) Construct an original and logically structured concept map that identifies system capabilities and their influence on the selection and effectiveness of vision-based applications, establishes clear interconnections, integrates multiple literature perspectives, and maintains high clarity and visual organization.

